

EEE-2421: Electrical Drives and Instrumentation

Credit Hours - 2

Segment - 2

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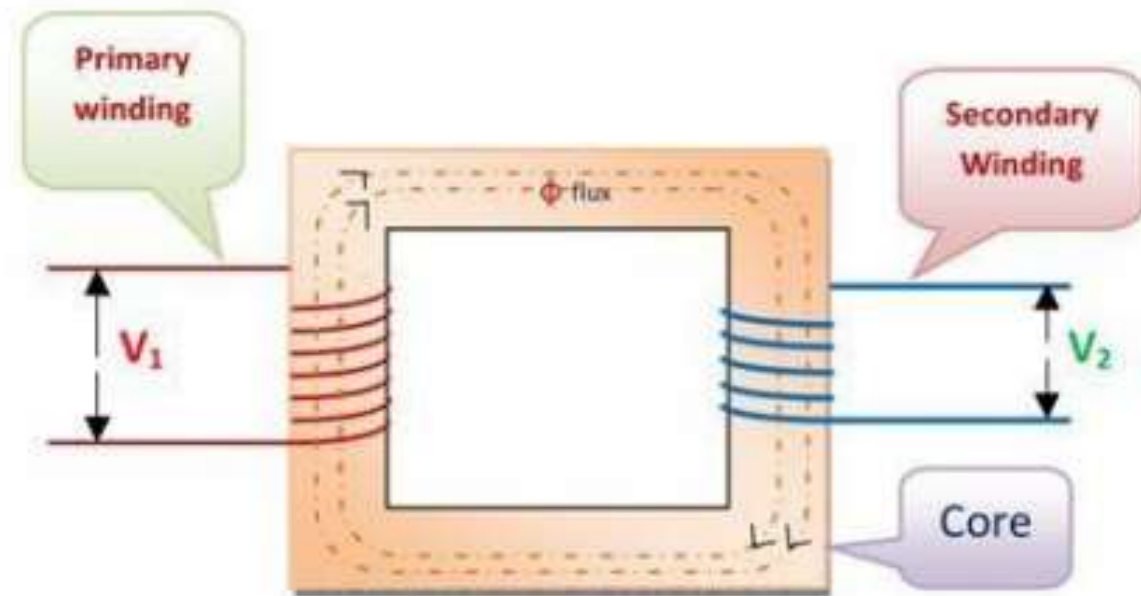
Reference Materials

- Principles of Electrical Machines by VK Mehta
- Numerical Problems from BL Thereja
- Youtube Channel [SAB DSE](#) Playlist [Electrical Drives and Instrumentation](#)

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What is Transformer?

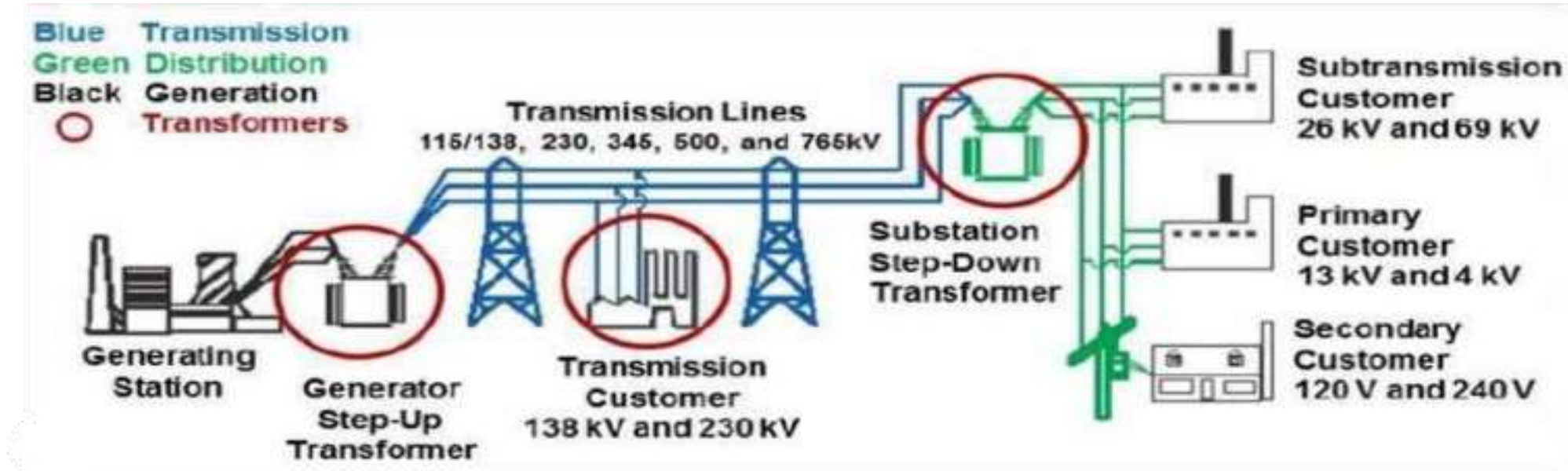
- Transformer is an electrical devices that changes the **magnitude of alternating voltage or current** from one value to another.
- It has **no moving parts** so it is a static device.



Significance of Transformer

Why Important ?

- Transformers are used to adjust the voltages to a proper level.
- It transfer electrical energy without changing the frequency.

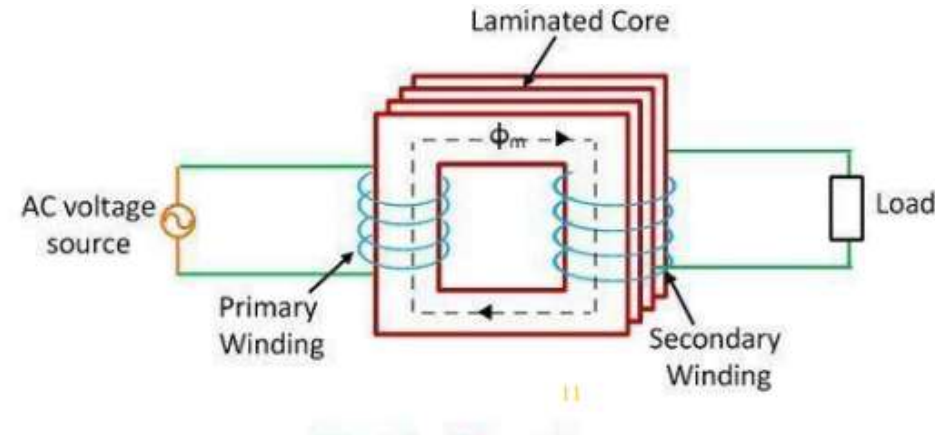


- Transformer is used to increase the voltage at the power generating station (Step up) to reduce the loss of the electricity in the transmission line and used to decrease the voltage (Step down) for house hold purpose. Because of very **rugged** and **durable in construction** it requires very little maintenance. Thus transformers are the basic components for electricity transmission.

Transformer Construction

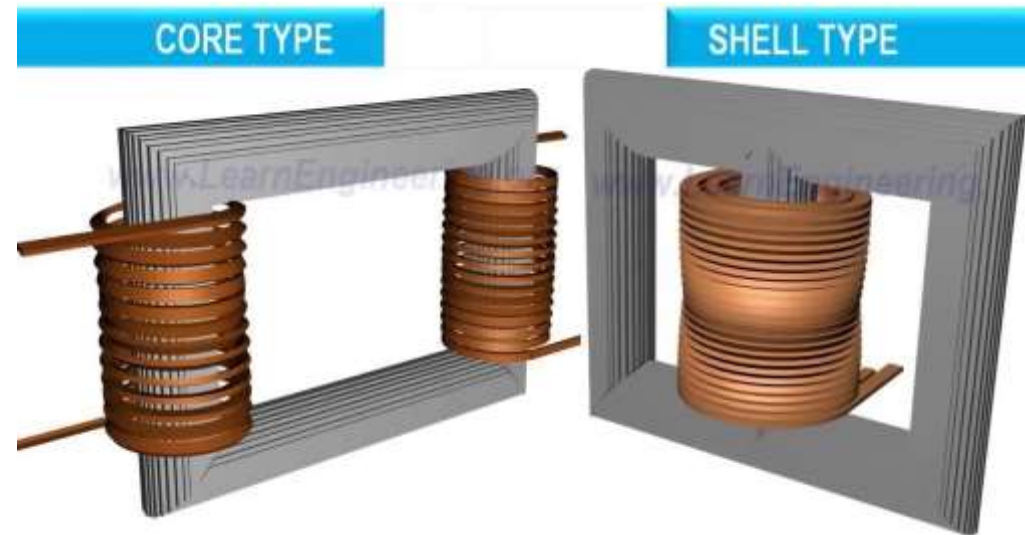
Transformers have mainly three parts—

1. Core
2. Primary winding and
3. Secondary winding



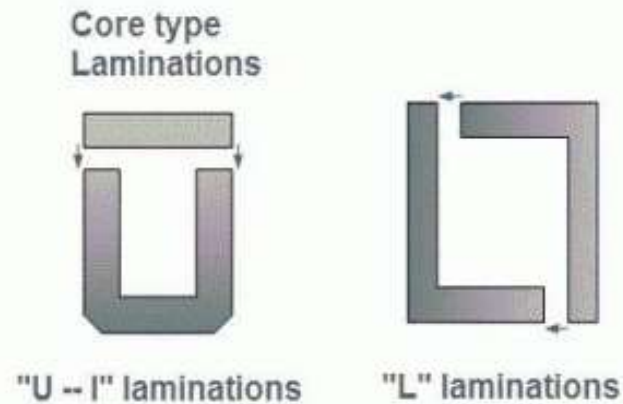
The transformer are two types according to core-

1. Core type.
2. Shell type.

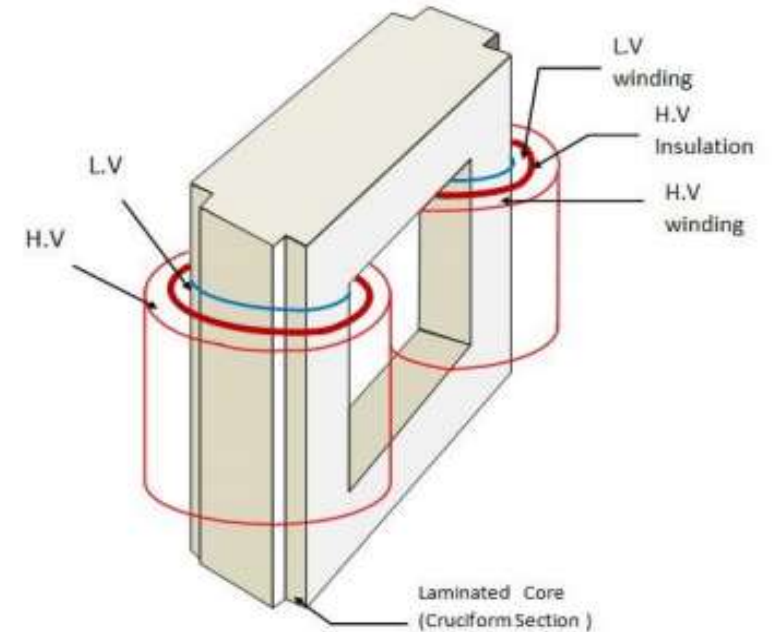


Core Type Transformer

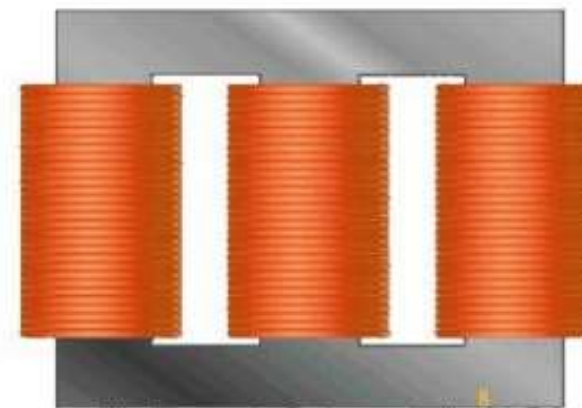
- ▶ In Core type transformer, the windings surround a considerable part of the core.
- ▶ In the core type transformer magnetic core is made from 'L' shape strips.



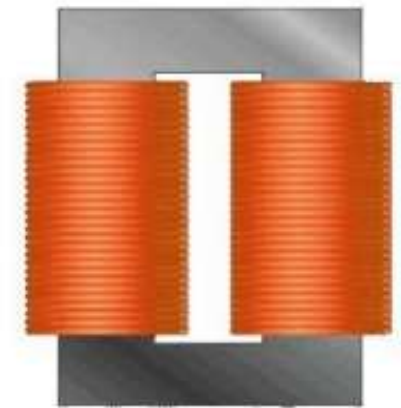
- ▶ For reducing leakage flux high voltage and low voltage winding are placed one after another separating with high insulation.



Core Type Transformer



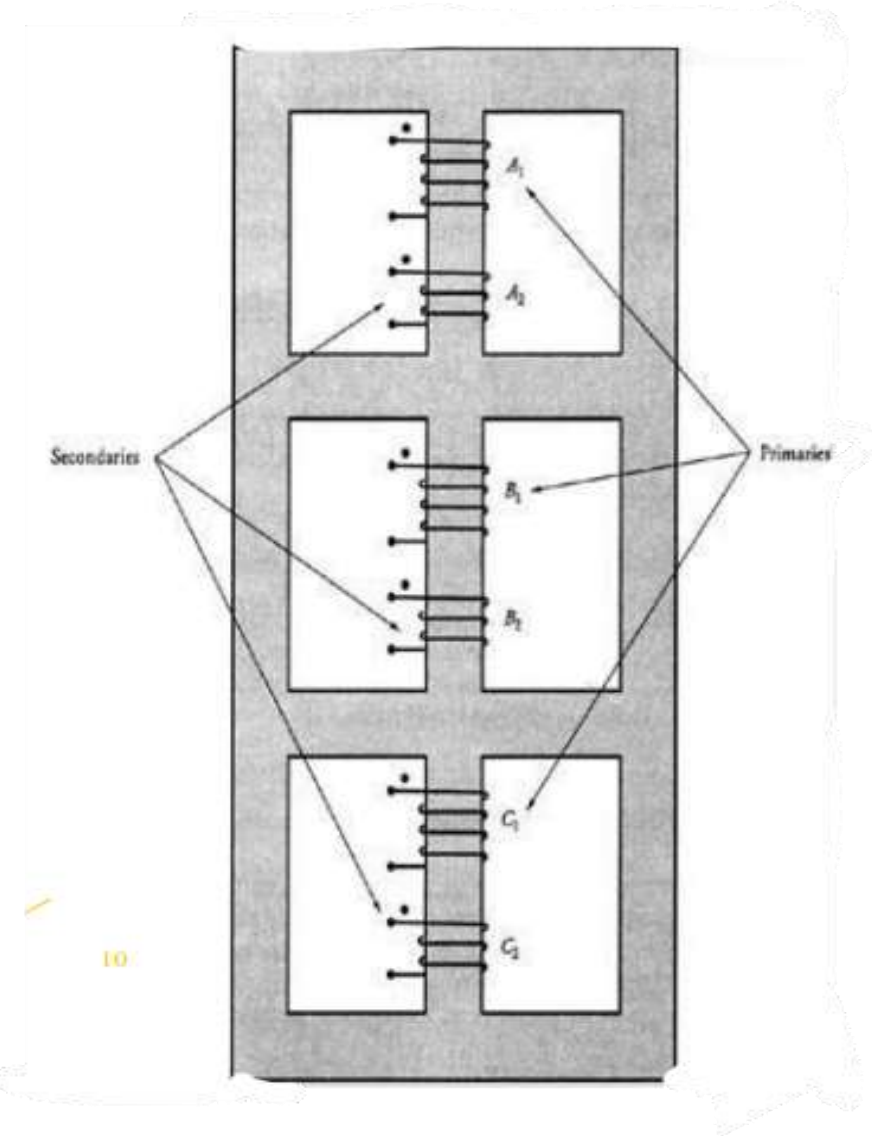
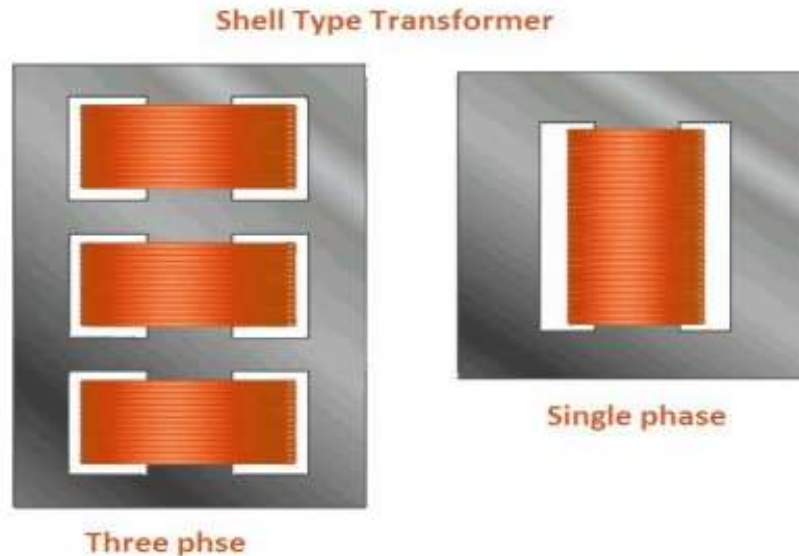
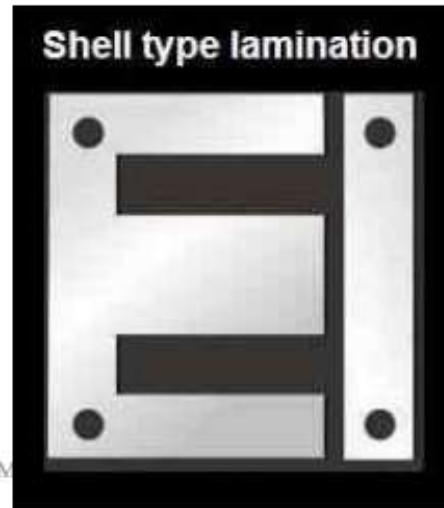
Three phase



Single phase

Shell Type Transformer

- ▶ In Shell type transformer, the core surround a considerable part of the winding.
- ▶ The core of shell type transformer is made from 'E' and 'I' strips.
- ▶ The width of the central limb is more than that of the outer limbs carries the whole of the flux.



Working Principle of Transformer

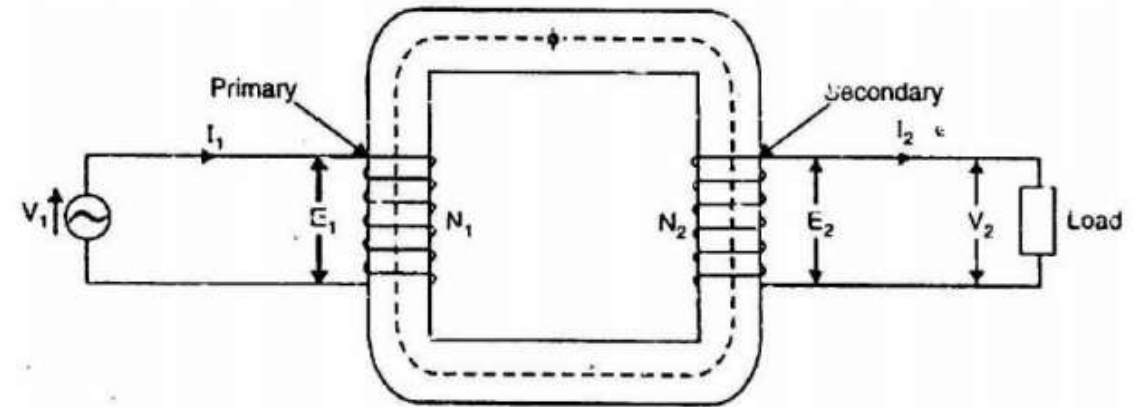
- When an alternating voltage V_1 is applied to the primary, an alternating flux ϕ is set up in the core. This alternating flux links both the windings and induces e.m.f.s E_1 and E_2 in them according to Faraday's laws of electromagnetic induction. The e.m.f. E_1 is termed as primary e.m.f. and e.m.f. E_2 is termed as secondary e.m.f.

- When V_1 is applied to the primary and **e.m.f. E_1** is induced known as **primary e.m.f.**

$$E_1 = -N_1 \frac{d\phi}{dt}$$

- The **e.m.f. E_2** is termed as **secondary e.m.f.**

$$E_2 = -N_2 \frac{d\phi}{dt}$$



- Note that magnitudes of E_2 and E_1 depend upon the number of turns on the secondary and primary respectively. If $N_2 > N_1$, then $E_2 > E_1$ (or $V_2 > V_1$) and **we get a step-up transformer**. On the other hand, if $N_2 < N_1$, then $E_2 < E_1$ (or $V_2 < V_1$) and **we get a step-down transformer**.
- If load is connected across the secondary winding, the secondary e.m.f. E_2 will cause a current I_2 to flow through the load. Thus, a transformer enables us to transfer a.c. power from one circuit to another with a change in voltage level.

Working Principle of Transformer

Transformer works on the principle of **Faraday's Electromagnetic Mutual Induction**.

Faradays First Law

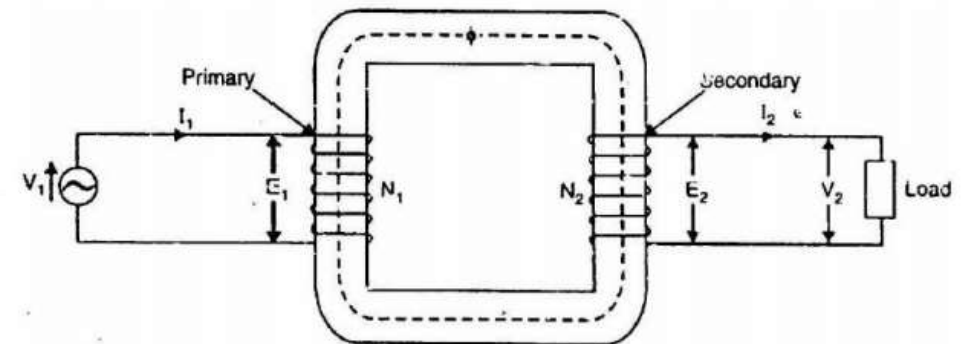
"Whenever a conductor is placed in a varying magnetic field, an electromotive force (emf)/(voltage) is induced. If the conductor circuit is closed, a current is induced which is called induced current."

Faradays Second Law

"The induced e.m.f. in a coil is equal to the rate of change of flux linkage."

$$E = N \frac{d\Phi}{dt}$$

Faradays Law of Induction



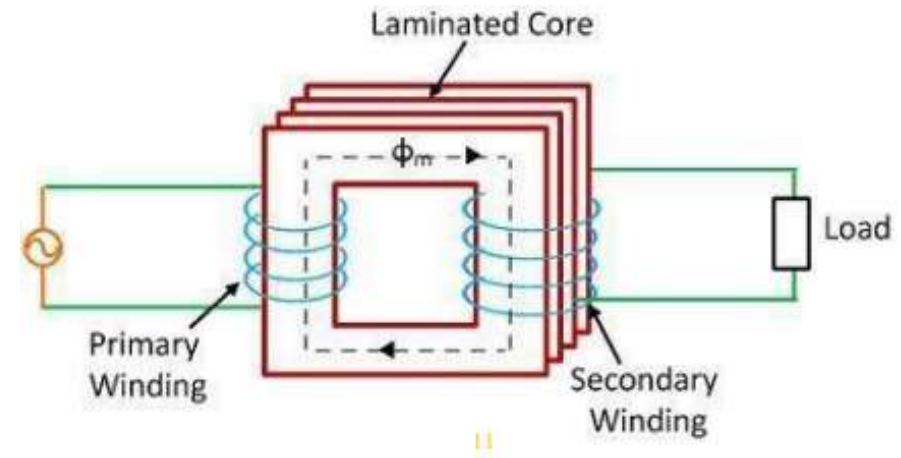
Transformer Basic Formulae

- ✓ Rms voltage induced in the primary winding

$$E_1 = 4.44 f N_1 \Phi_m = 4.44 f N_1 B_m A$$

- ✓ Rms voltage induced in the secondary winding

$$E_2 = 4.44 f N_2 \Phi_m = 4.44 f N_2 B_m A$$



From equations (i) and (ii), we get

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

$$a = \frac{V_1}{V_2} = \frac{N_1}{N_2}$$

a = Turn Ratio

K = Voltage Transformation Ratio

This constant K is known as voltage transformation ratio.

(i) If $N_2 > N_1$ i.e. $K > 1$, then transformer is called **step-up** transformer.

(ii) If $N_2 < N_1$ i.e. $K < 1$, then transformer is known as **step-down** transformer.

Again, for an *ideal* transformer, input VA = output VA.

$$V_1 I_1 = V_2 I_2 \text{ or } \frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{1}{K}$$

Transformer Numerical Problems

Example 32.1. The maximum flux density in the core of a 250/3000-volts, 50-Hz single-phase transformer is 1.2 Wb/m^2 . If the e.m.f. per turn is 8 volt, determine

(i) primary and secondary turns (ii) area of the core.

Solution. (i)

$$E_1 = N_1 \times \text{e.m.f. induced/turn}$$

$$N_1 = 250/8 = 32; N_2 = 3000/8 = 375$$

(ii) We may use

$$E_2 = 4.44 f N_2 B_m A$$

$\therefore$

$$3000 = 4.44 \times 50 \times 375 \times 1.2 \times A; A = 0.03 \text{ m}^2.$$

$$E_1 = 4.44 f N_1 \Phi_m = 4.44 f N_1 B_m A$$

$$E_2 = 4.44 f N_2 \Phi_m = 4.44 f N_2 B_m A$$

Example 32.2. The core of a 100-kVA, 11000/550 V, 50-Hz, 1-ph, core type transformer has a cross-section of $20 \text{ cm} \times 20 \text{ cm}$. Find (i) the number of H.V. and L.V. turns per phase and (ii) the e.m.f. per turn if the maximum core density is not to exceed 1.3 Tesla. Assume a stacking factor of 0.9.

Solution. (i)

$$B_m = 1.3 \text{ T}, A = (0.2 \times 0.2) \times 0.9 = 0.036 \text{ m}^2$$

$\therefore$

$$11,000 = 4.44 \times 50 \times N_1 \times 1.3 \times 0.036, N_1 = 1060$$

$$550 = 4.44 \times 50 \times N_2 \times 1.3 \times 0.036; N_2 = 53$$

or,

$$N_2 = K N_1 = (550/11,000) \times 1060 = 53$$

(ii)

$$\text{e.m.f./turn} = 11,000/1060 = 10.4 \text{ V or } 550/53 = 10.4 \text{ V}$$

Transformer Numerical Problems

Example 32.3. A single-phase transformer has 400 primary and 1000 secondary turns. The net cross-sectional area of the core is 60 cm^2 . If the primary winding be connected to a 50-Hz supply at 520 V, calculate (i) the peak value of flux density in the core (ii) the voltage induced in the secondary winding.

Solution.

$$K = N_2/N_1 = 1000/400 = 2.5$$

(i) $E_2/E_1 = K \therefore E_2 = KE_1 = 2.5 \times 520 = \mathbf{1300 \text{ V}}$

(ii) $E_1 = 4.44 f N_1 B_m A$

or $520 = 4.44 \times 50 \times 400 \times B_m \times (60 \times 10^{-4}) \therefore B_m = \mathbf{0.976 \text{ Wb/m}^2}$

From equations (i) and (ii), we get

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

Example 32.4. A 25-kVA transformer has 500 turns on the primary and 50 turns on the secondary winding. The primary is connected to 3000-V, 50-Hz supply. Find the full-load primary and secondary currents, the secondary e.m.f. and the maximum flux in the core. Neglect leakage drops and no-load primary current.

Solution.

$$K = N_2/N_1 = 50/500 = 1/10$$

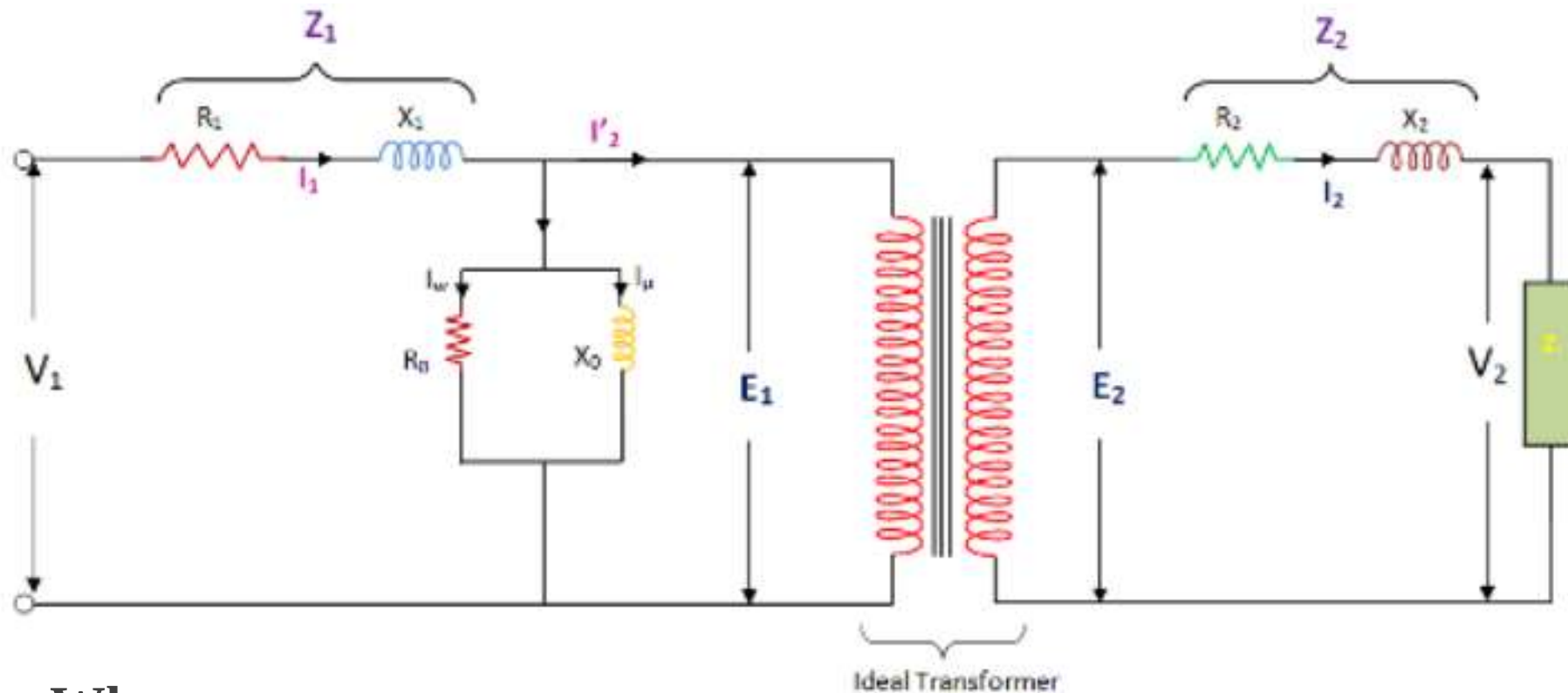
Now, full-load $I_1 = 25,000/3000 = \mathbf{8.33 \text{ A}}$. F.L. $I_2 = I_1/K = 10 \times 8.33 = \mathbf{83.3 \text{ A}}$

e.m.f. per turn on primary side = $3000/500 = 6 \text{ V}$

$\therefore$ secondary e.m.f. = $6 \times 50 = \mathbf{300 \text{ V}}$ (or $E_2 = KE_1 = 3000 \times 1/10 = 300 \text{ V}$)

Also, $E_1 = 4.44 f N_1 \Phi_m$; $3000 = 4.44 \times 50 \times 500 \times \Phi_m \therefore \Phi_m = \mathbf{27 \text{ mWb}}$

Practical Transformer Equivalent Circuit



Where,

R_1 = Primary Winding Resistance

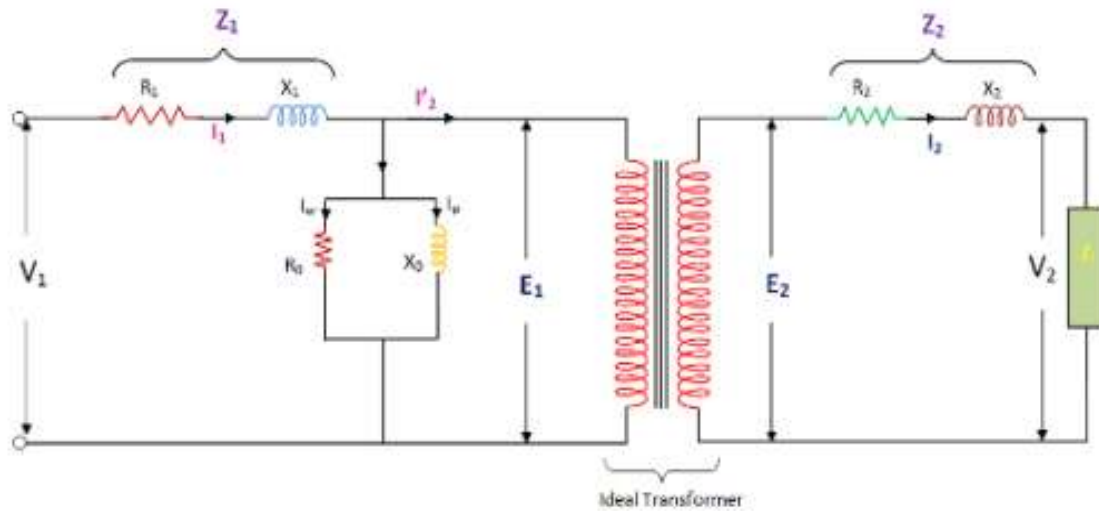
R_2 = Secondary winding Resistance

I_0 = No-load current

I_μ = Magnetizing Component

I_w = Working Component

Equivalent Circuit Referred to Primary



****Equivalent Circuit diagram of Transformer****

Where,

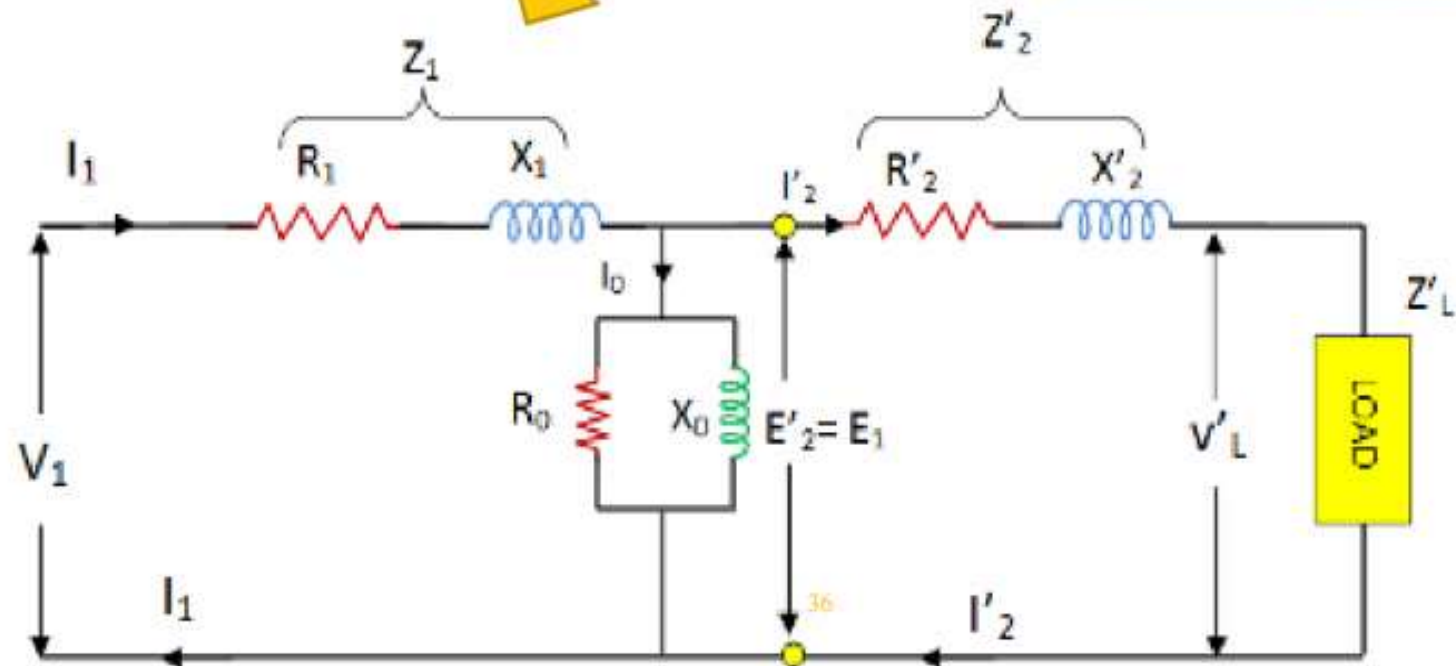
R_1 = Primary Winding Resistance

R_2 = Secondary winding Resistance

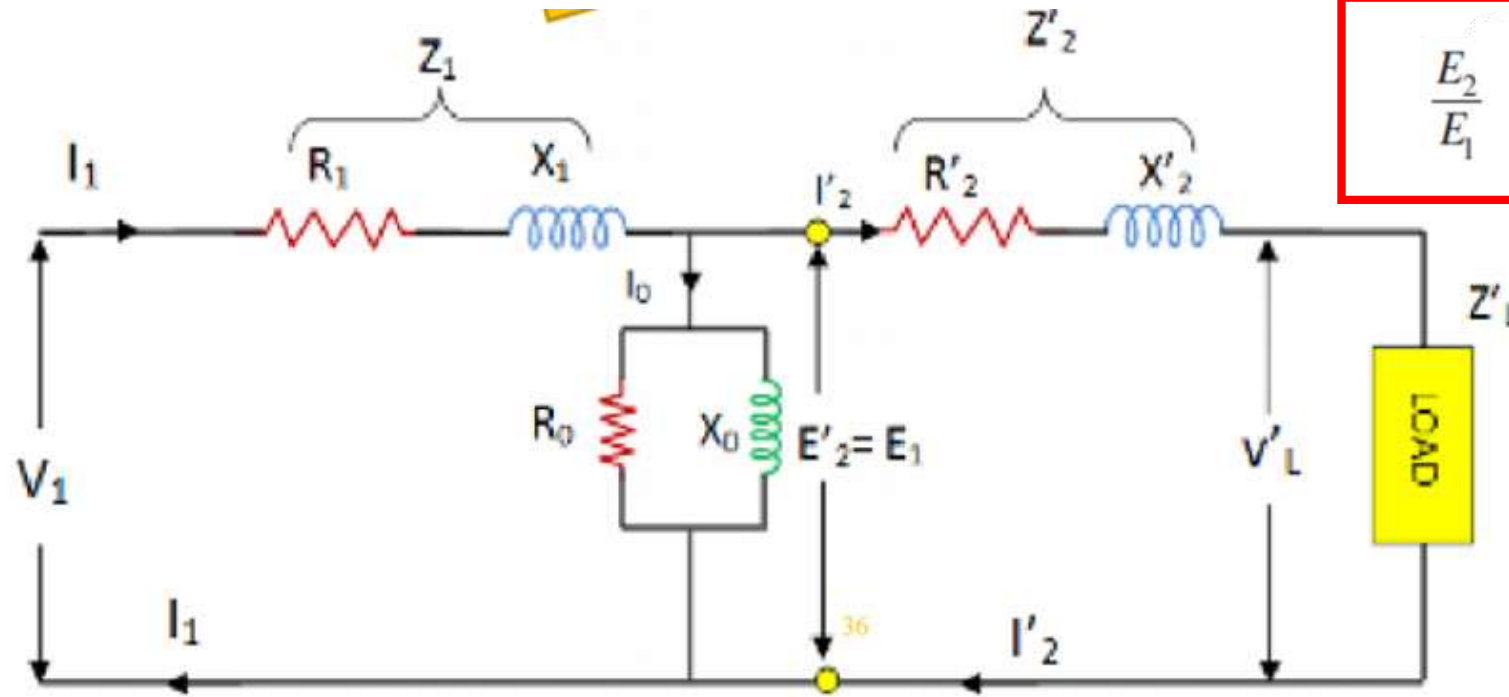
I_0 = No-load current

I_μ = Magnetizing Component

I_w = Working Component



Equivalent Circuit Referred to Primary



$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K \quad a = \frac{V_1}{V_2} = \frac{N_1}{N_2}$$

$$R'_2 = \frac{V_1}{I_1} = \frac{V_2/K}{K I_2} = \frac{V_2}{I_2 K^2} = \frac{R_2}{K^2}$$

$$R'_2 = R_2/K^2$$

$$X'_2 = X_2/K^2$$

$$Z'_2 = Z_2/K^2$$

$$Z'_L = \frac{V_P}{I_P} = \frac{a V_S}{I_S/a} = a^2 \frac{V_S}{I_S}$$

$$Z'_L = a^2 Z_L$$

Where,

R_1 = Primary Winding Resistance

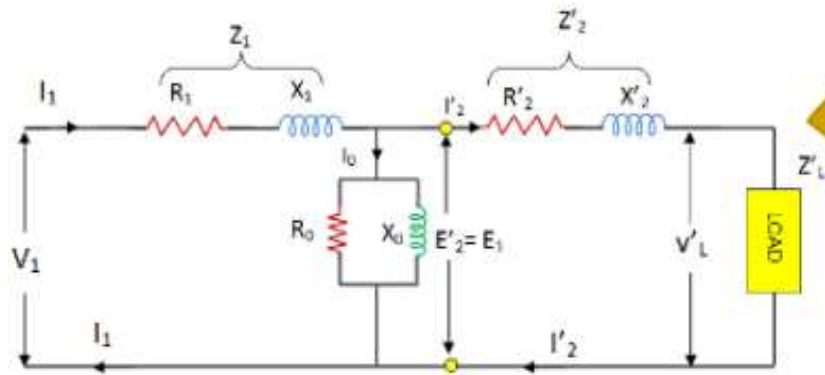
R_2 = Secondary winding Resistance

R'_2 = Secondary Resistance Referred to Primary

a = Turn Ratio

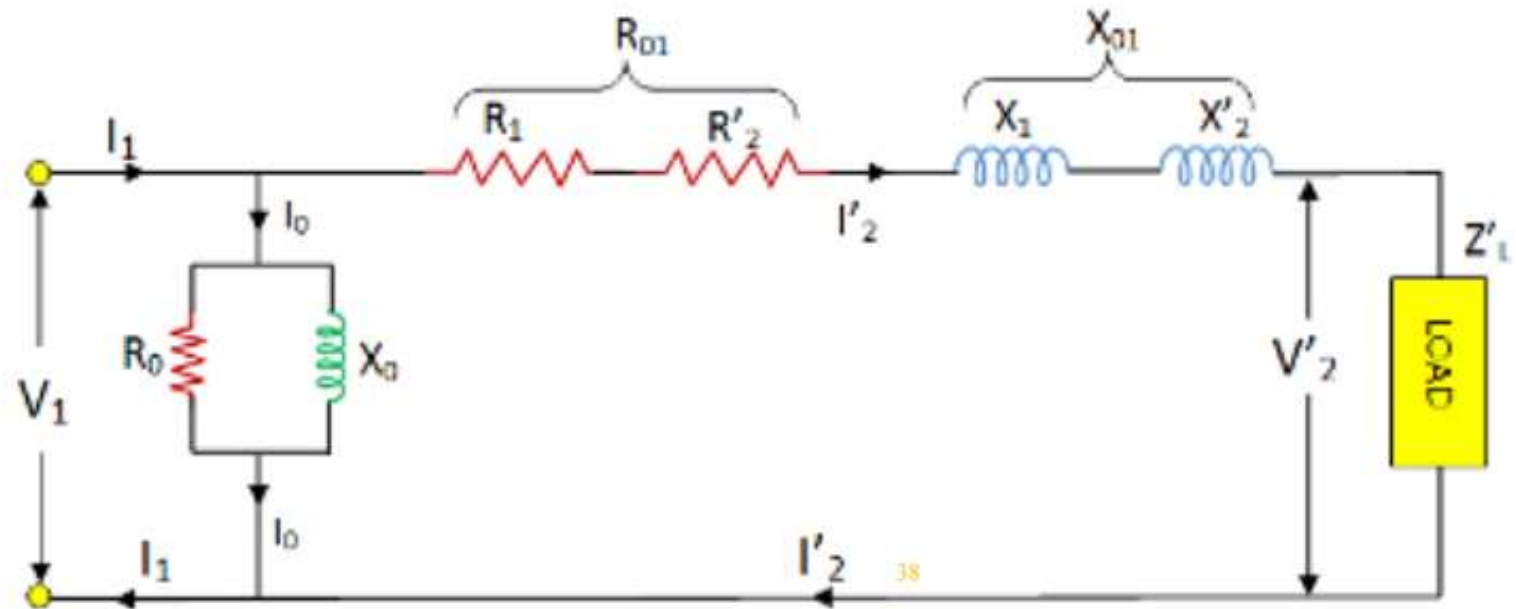
K = Voltage Transformation Ratio

Exact Equivalent Circuit



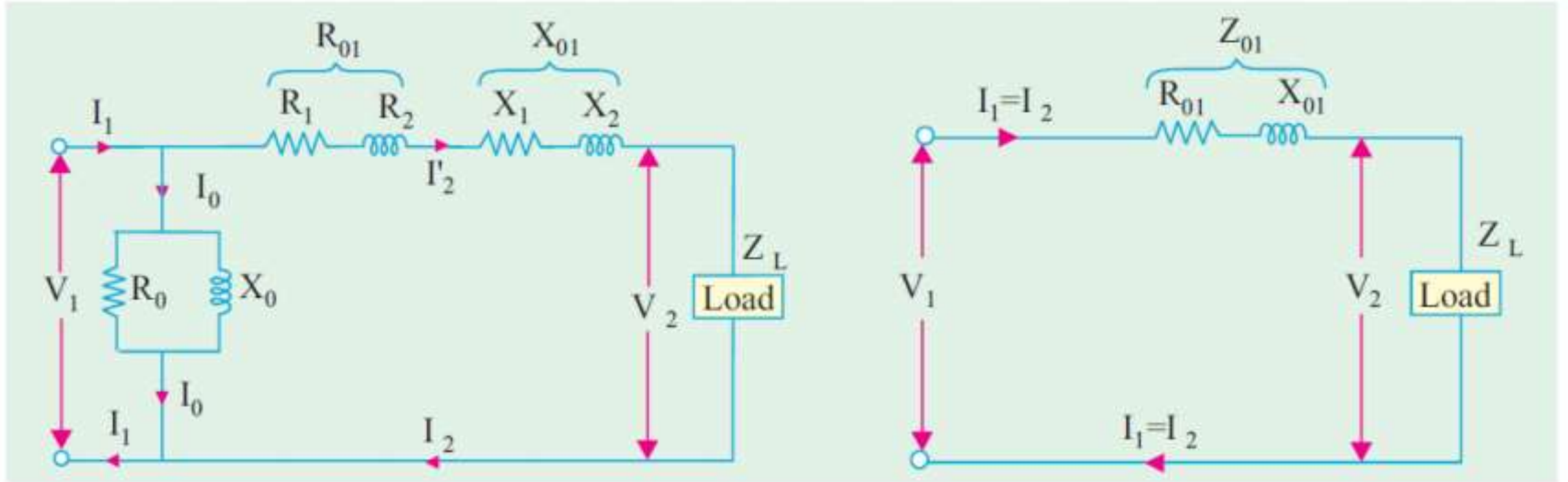
Exact Equivalent circuit:

A simplification can be made by transferring the *exciting circuit* across the terminals.



Approximate Equivalent Circuit

At last, the circuit is simplified by omitting I_0 altogether as shown in figure below



Shifting Impedance

Shifting Impedance by Voltage Transformation Ratio

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

The primary equivalent of the secondary induced voltage is $E_2' = E_2/K = E_1$.

Similarly, primary equivalent of secondary terminal or output voltage is $V_2' = V_2/K$.

Primary equivalent of the secondary current is $I_2' = KI_2$.

For transferring secondary impedance to primary K^2 is used.

$$R_2' = R_2/K^2$$

$$X_2' = X_2/K^2$$

$$Z_2' = Z_2/K^2$$

Shifting Impedance by Turn Ratio

- ▶ The *impedance* of a device is defined:

$$Z_L = \frac{V_L}{I_L}$$

- ▶ Transformer changes the *ratio* between voltage and current and hence the apparent impedance of an element.

$$Z_L = \frac{V_S}{I_S}$$

- ▶ The impedance of the load is given by :

- ▶ The apparent impedance of the primary circuit is:

$$Z_L' = \frac{V_P}{I_P}$$

$$Z_L' = \frac{V_P}{I_P} = \frac{aV_S}{I_S/a} = a^2 \frac{V_S}{I_S}$$

$$Z_L' = a^2 Z_L$$

Transformer Numerical Problems

Example 32.15. A 30 kVA, 2400/120-V, 50-Hz transformer has a high voltage winding resistance of 0.1Ω and a leakage reactance of 0.22Ω . The low voltage winding resistance is 0.035Ω and the leakage reactance is 0.012Ω . Find the equivalent winding resistance, reactance and impedance referred to the (i) high voltage side and (ii) the low-voltage side.

(Electrical Machines-I, Bangalore Univ. 1987)

Solution.

$$K = 120/2400 = 1/20; R_1 = 0.1 \Omega, X_1 = 0.22 \Omega$$

$$R_2 = 0.035 \Omega \quad \text{and} \quad X_2 = 0.012 \Omega$$

(i) Here, high-voltage side is, obviously, the primary side. Hence, values as referred to primary side are

$$R_{01} = R_1 + R_2' = R_1 + R_2/K^2 = 0.1 + 0.035/(1/20)^2 = 14.1 \Omega$$

$$X_{01} = X_1 + X_2' = X_1 + X_2/K^2 = 0.22 + 0.012/(1/20)^2 = 5.02 \Omega$$

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2} = \sqrt{14.1^2 + 5.02^2} = 15 \Omega$$

(ii)

$$R_{02} = R_2 + R_1' = R_2 + K^2 R_1 = 0.035 + (1/20)^2 \times 0.1 = 0.03525 \Omega$$

$$X_{02} = X_2 + X_1' = X_2 + K^2 X_1 = 0.012 + (1/20)^2 \times 0.22 = 0.01255 \Omega$$

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2} = \sqrt{0.0325^2 + 0.01255^2} = 0.0374 \Omega$$

$$(\text{or } Z_{02} = K^2 Z_{01} = (1/20)^2 \times 15 = 0.0375 \Omega)$$

Losses in Transformer

- Copper Loss

The **resistive heating losses** in the primary and secondary windings of the transformer.

To reduce the loss **size of the winding** conductor is selected very carefully.

- This loss is proportional to the **square of the current** in the windings.
- To determine the copper losses, **short circuit test** is performed.
- The total copper losses in Transformer are:

$$= I_1^2 R_1 + I_2^2 R_2$$

- Iron losses

The power losses that take place in its iron core

These losses occur due to alternating flux set up in the core.

To determine the iron losses, open circuit test of transformer is performed.

Iron loss is two types

- *Eddy current losses*
- *Hysteresis losses*

Transformer Numerical Problems

Example 32.16. A 50-kVA, 4,400/220-V transformer has $R_1 = 3.45 \Omega$, $R_2 = 0.009 \Omega$. The values of reactances are $X_1 = 5.2 \Omega$ and $X_2 = 0.015 \Omega$. Calculate for the transformer (i) equivalent resistance as referred to primary (ii) equivalent resistance as referred to secondary (iii) equivalent reactance as referred to both primary and secondary (iv) equivalent impedance as referred to both primary and secondary (v) total Cu loss, first using individual resistances of the two windings and secondly, using equivalent resistances as referred to each side.

Solution. Full-load $I_1 = 50,000/4,400 = 11.36 \text{ A}$ (assuming 100% efficiency)

Full-load $I_2 = 50,000/220 = 227 \text{ A}$; $K = 220/4,400 = 1/20$

(i) $R_{01} = R_1 + \frac{R_2}{K^2} = 3.45 + \frac{0.009}{(1/20)^2} = 3.45 + 3.6 = 7.05 \Omega$

(ii) $R_{02} = R_2 + K^2 R_1 = 0.009 + (1/20)^2 \times 3.45 = 0.009 + 0.0086 = 0.0176 \Omega$

Also, $R_{02} = K^2 R_{01} = (1/20)^2 \times 7.05 = 0.0176 \Omega$ (check)

(iii) $X_{01} = X_1 + X_2' = X_1 + X_2/K^2 = 5.2 + 0.015/(1/20)^2 = 11.2 \Omega$

$X_{02} = X_2 + X_1' = X_2 + K^2 X_1 = 0.015 + 5.2/20^2 = 0.028 \Omega$

Also $X_{02} = K^2 X_{01} = 11.2/400 = 0.028 \Omega$ (check)

(iv) $Z_{01} = \sqrt{(R_{01}^2 + X_{01}^2)} = \sqrt{(7.05^2 + 11.2^2)} = 13.23 \Omega$

$Z_{02} = \sqrt{(R_{02}^2 + X_{02}^2)} = \sqrt{(0.0176^2 + 0.028^2)} = 0.03311 \Omega$

Also $Z_{02} = K^2 Z_{01} = 13.23/400 = 0.0331 \Omega$ (check)

(v) Cu loss $= I_1^2 R_1 + I_2^2 R_2 = 11.36^2 \times 3.45 + 227^2 \times 0.009 = 910 \text{ W}$

Also Cu loss $= I_1^2 R_{01} = 11.36^2 \times 7.05 = 910 \text{ W}$

$= I_2^2 R_{02} = 227^2 \times 0.0176 = 910 \text{ W}$

The total copper losses in Transformer are:

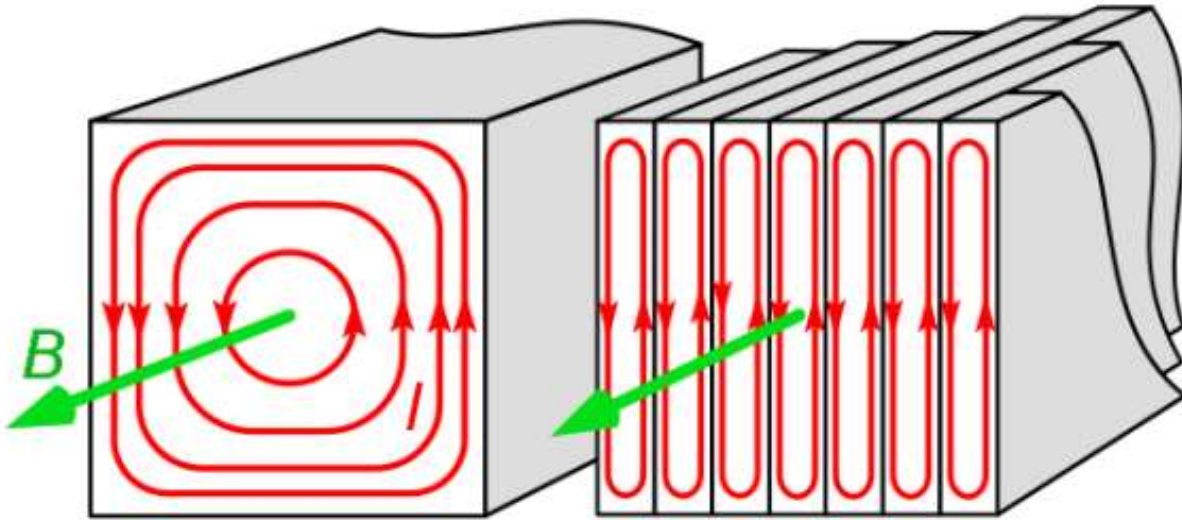
$$= I_1^2 R_1 + I_2^2 R_2$$

Losses in Transformer

- Eddy Current Loss

Because of time variation of flux in the magnetic material, current is induced in the magnetic material. This current is called Eddy current which causes loss.

- They are proportional to the square of the voltage applied to the transformer.
- This current is reduced by using laminated (thin sheet) iron structure, with insulation between the laminations.



$$\text{Eddy Current Loss } (P_e) = K_e B_m^2 f^2 t^2 V$$

Where,

P_e = Eddy current loss (Watt)

B_m = Maximum Flux density (Wb/m^2)

f = Frequency of supply (f)

t = Thickness of lamination (meter)

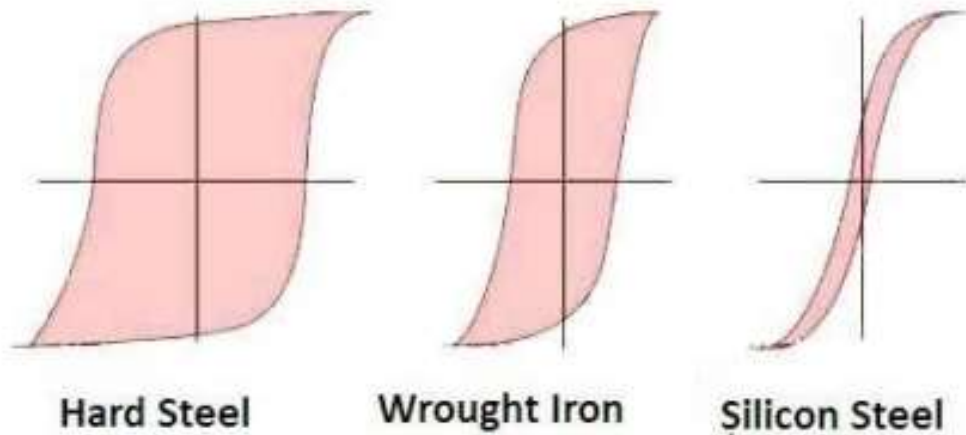
V = Volume of the material (M^3)

K_e = Eddy current Constant

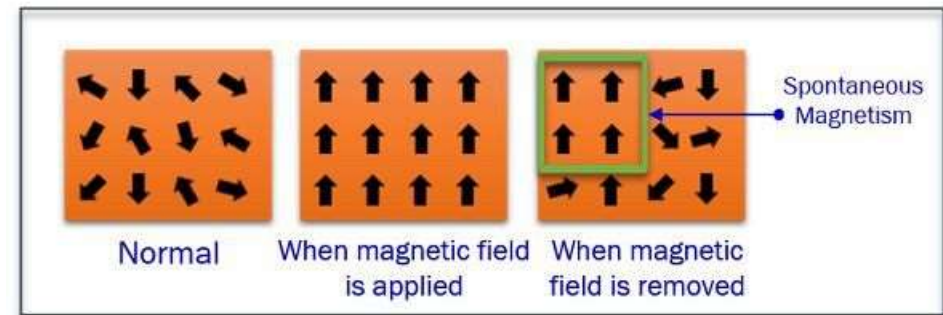
Losses in Transformer

- Hysteresis Loss

- Hysteresis losses are associated with the rearrangement of the magnetic domains.
- When a magnetic material is subjected to alternating flux, power is required for the continuous reversal of molecular magnets.
- This power is dissipated in the form of heat and is known as 'Hysteresis Loss'. To reduce loss we need to select such type of material which has small hysteresis area. **That's why Silicon Steel is used in transformer cores.**



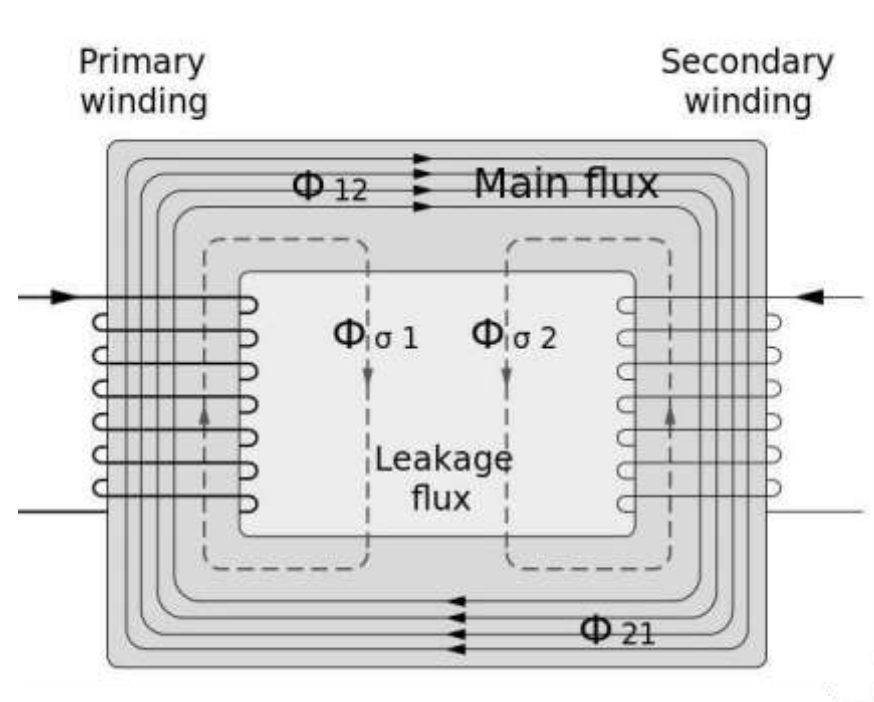
$$\text{Hysteresis loss} = P_h = \eta B_{max}^{1.6} fV$$



Ferromagnetic Material

Leakage Flux

- The fluxes which escape the core and pass through only one of the transformer windings are leakage fluxes.
- These escaped fluxes produce a self-inductance in the primary and secondary coils.



Why is transformer rated in KVA?

Transformer is rated in VA. **But why?**

- As **copper loss** depends on current and **core loss** depends on voltage.
- Also we do not know the value or nature of load so we don't know **power factor**.
- So transformer rated in **VA, KVA, MVA** instead of **KW or MW**.

Voltage Regulation and Efficiency

The voltage regulation of a transformer is the arithmetic difference between the ***no-load*** secondary voltage (V_2) and the secondary voltage V_2 ***on load*** expressed as percentage of no-load voltage.

$$\% \text{age voltage regulation} = \frac{V_2 - V_2}{V_2} \times 100$$

$$\text{Efficiency} = \frac{\text{Output}}{\text{Input}}$$

$$\text{Efficiency} = \frac{\text{Output}}{\text{Output} + \text{losses}} = \frac{\text{Output}}{\text{Output} + \text{Cu loss} + \text{iron loss}}$$

$$\eta = \frac{\text{Input} - \text{Losses}}{\text{Input}} = 1 - \frac{\text{losses}}{\text{Input}}$$

Conditions for Maximum Efficiency

$$\text{Cu loss} = I_1^2 R_{01} \quad \text{or} \quad I_2^2 R_{02} = W_{cu}$$

$$\text{Iron loss} = \text{Hysteresis loss} + \text{Eddy current loss} = W_h + W_e = W_i$$

Considering primary side,

$$\text{Primary input} = V_1 I_1 \cos \phi_1$$

$$\begin{aligned} \eta &= \frac{V_1 I_1 \cos \phi_1 - \text{losses}}{V_1 I_1 \cos \phi_1} = \frac{V_1 I_1 \cos \phi_1 - I_1^2 R_{01} - W_i}{V_1 I_1 \cos \phi_1} \\ &= 1 - \frac{I_1 R_{01}}{V_1 \cos \phi_1} - \frac{W_i}{V_1 I_1 \cos \phi_1} \end{aligned}$$

Differentiating both sides with respect to I_1 , we get

$$\frac{d\eta}{dI_1} = 0 - \frac{R_{01}}{V_1 \cos \phi_1} + \frac{W_i}{V_1 I_1^2 \cos \phi_1}$$

For η to be maximum, $\frac{d\eta}{dI_1} = 0$. Hence, the above equation becomes

$$\frac{R_{01}}{V_1 \cos \phi_1} = \frac{W_i}{V_1 I_1^2 \cos \phi_1} \quad \text{or} \quad W_i = I_1^2 R_{01} \quad \text{or} \quad I_2^2 R_{02}$$

or

$$\text{Cu loss} = \text{Iron loss}$$

Transformer Numerical Problems

Example 32.21. A 230/460-V transformer has a primary resistance of $0.2\ \Omega$ and reactance of $0.5\ \Omega$ and the corresponding values for the secondary are $0.75\ \Omega$ and $1.8\ \Omega$ respectively. Find the secondary terminal voltage when supplying 10 A at 0.8 p.f. lagging.

(Electric. Machines-II, Bangalore Univ. 1991)

Solution.

$$K = 460/230 = 2; R_{02} = R_2 + K^2 R_1 = 0.75 + 2^2 \times 0.2 = 1.55\ \Omega$$

$$X_{02} = X_2 + K^2 X_1 = 1.8 + 2^2 \times 0.5 = 3.8\ \Omega$$

$$\text{Voltage drop} = I_2 (R_{02} \cos \phi + X_{02} \sin \phi) = 10 (1.55 \times 0.8 + 3.8 \times 0.6) = 35.2\text{ V}$$

$$\therefore \text{Secondary terminal voltage} = 460 - 35.2 = \mathbf{424.8\text{ V}}$$

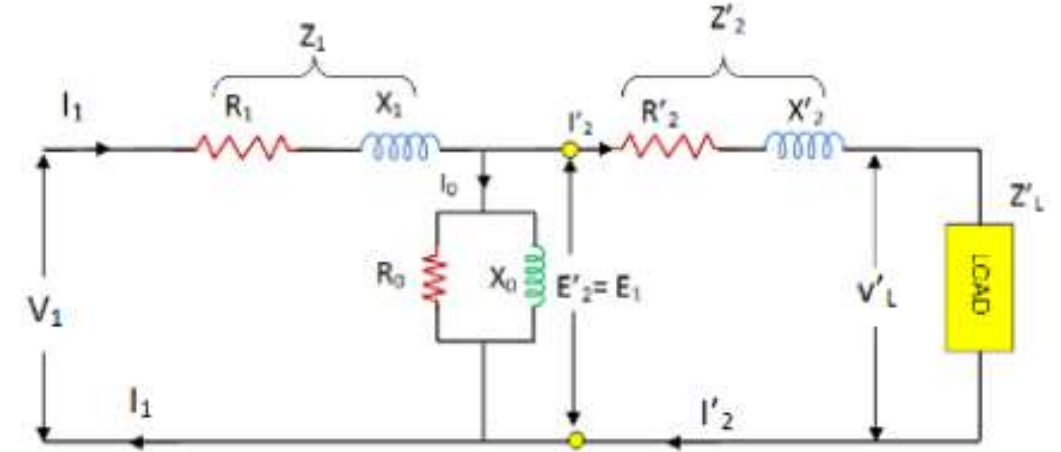
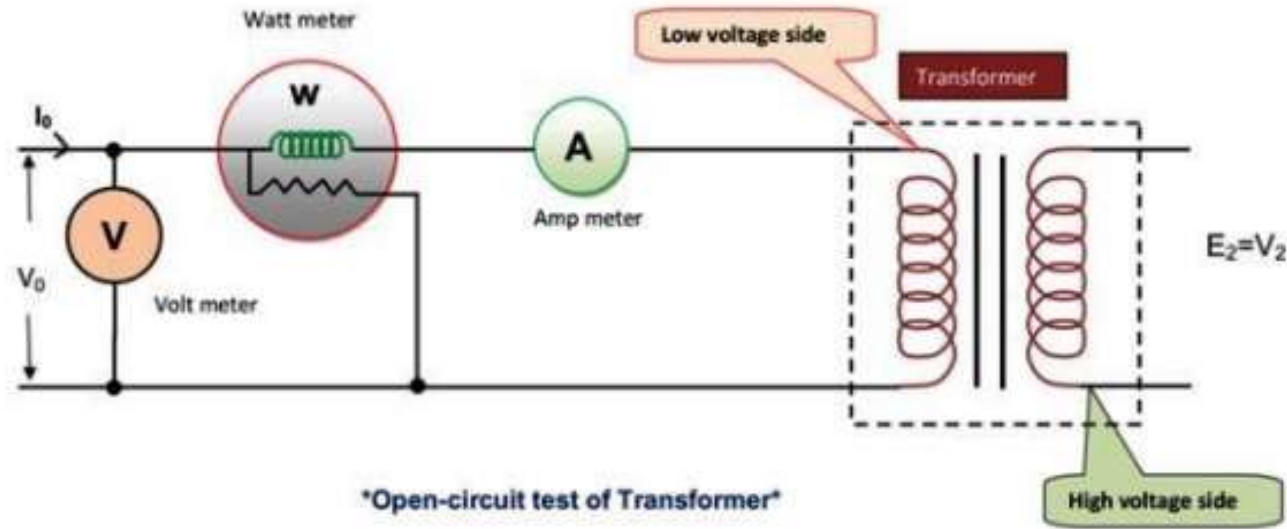
Transformer Tests

The performance of a transformer can be calculated on the basis of its equivalent circuit which contains four main parameters.

These Parameters can be easily determined by two tests:

- *Open circuit test or No load test*
- *Short circuit test or Impedance test*

No Load Test/Open Circuit Test



- Core losses of the transformer can be determined by open circuit test.
- The high voltage winding is left open.
- Normal voltage is applied to the low voltage winding.
- A wattmeter, a voltmeter and an ammeter are connected in the low voltage winding.
- **The Wattmeter measures the iron loss of the transformer. P_0**
- **The Ammeter measures the no load current I_0 which is very small.**
- **The voltmeter measures the applied voltage in the low voltage winding. V_0**

No Load Test/Open Circuit Test

Let

W is the wattmeter reading

V_1 is the applied voltage

I_0 is the ammeter reading

I_μ is the magnetizing component of no load current

I_w is the core loss component of no load current

Then

$$W = V_1 I_0 \cos \Phi_0$$

$$\therefore \text{No load p.f } \cos \Phi_0 = W / V_1 I_0$$

from the vector diagram of no-load transformer,

$$I_\mu = I_0 \sin \Phi_0,$$

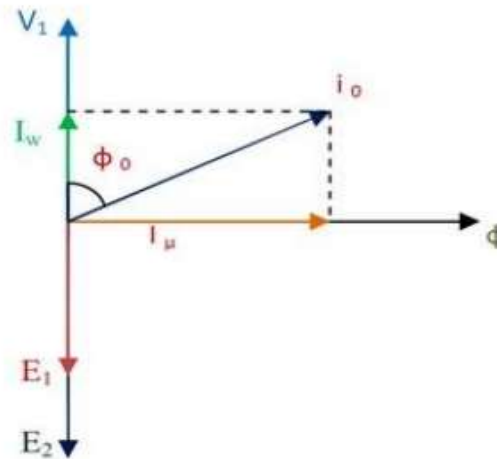
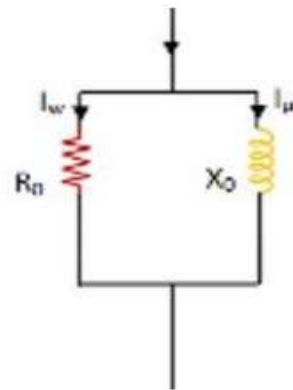
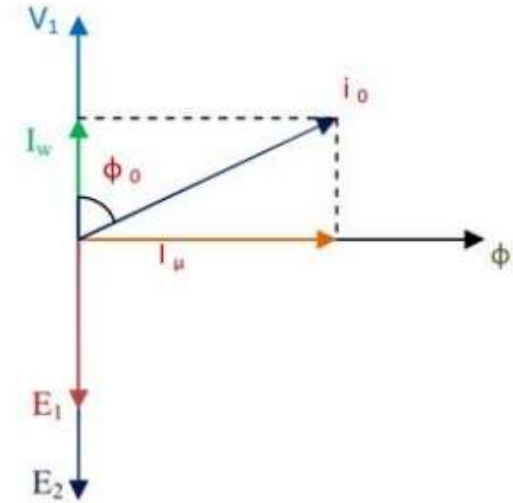
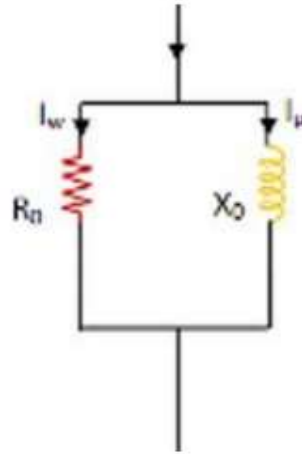
$$I_w = I_0 \cos \Phi_0,$$

$$\therefore X_0 = V_1 / I_\mu \text{ and}$$

$$R_0 = V_1 / I_w$$

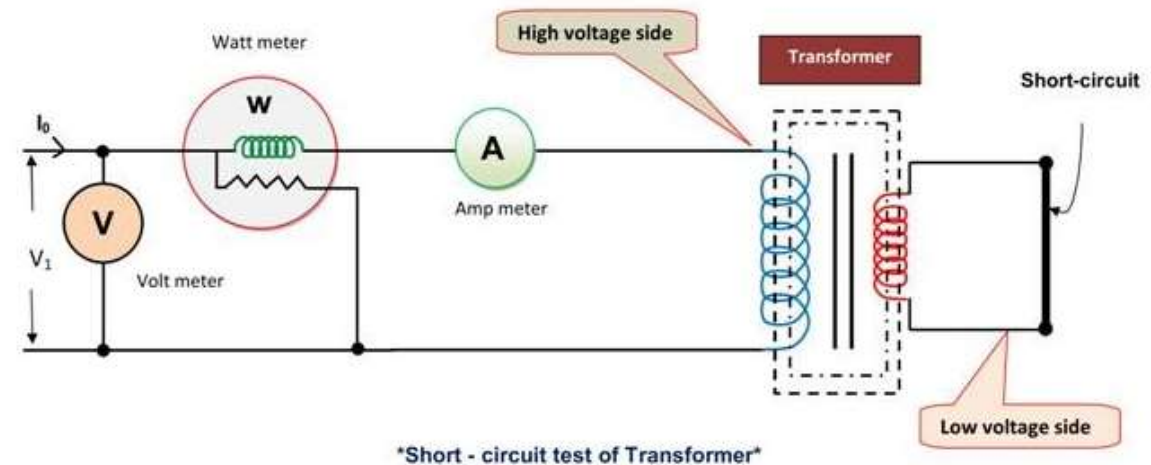
Thus,

$$Z_0 = \sqrt{R_0^2 + X_0^2}$$



Short Circuit Test

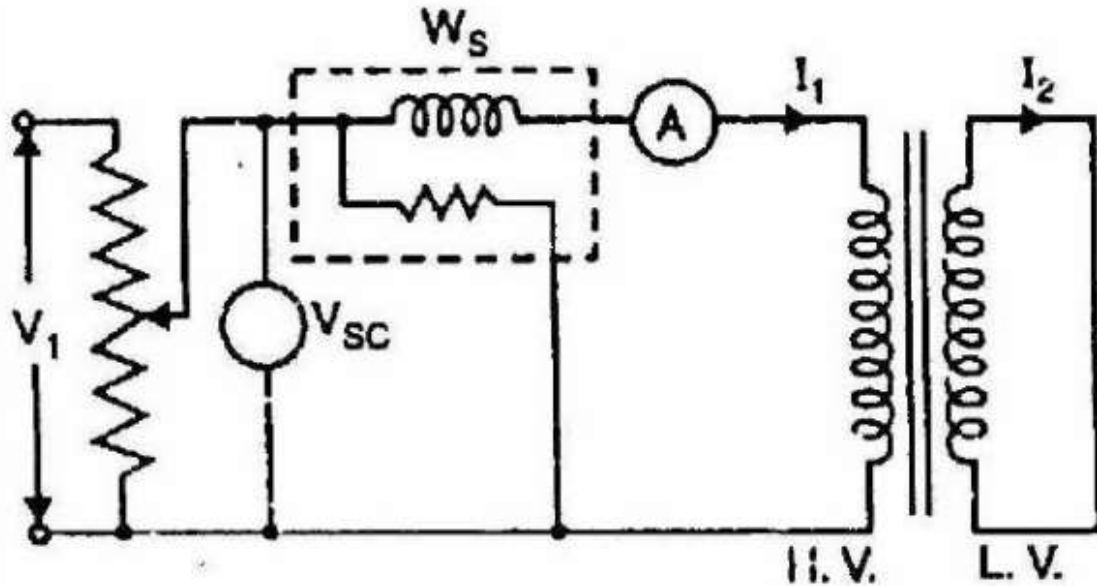
- This test is conducted to determine R_{01} (or R_{02}), X_{01} (or X_{02}) and full-load copper losses of the transformer.
- In this test, the low-voltage winding is short-circuited by a thick conductor and normal voltage is applied to the another winding.



- The applied voltage is slowly increased until the ammeter gives reading equal to the rated current of the HV side.
- When the current of HV side reach at rated current of the transformer, wattmeter represents the total full-load Cu-losses of the transformer i.e both primary Cu loss and secondary Cu loss.

Short Circuit Test

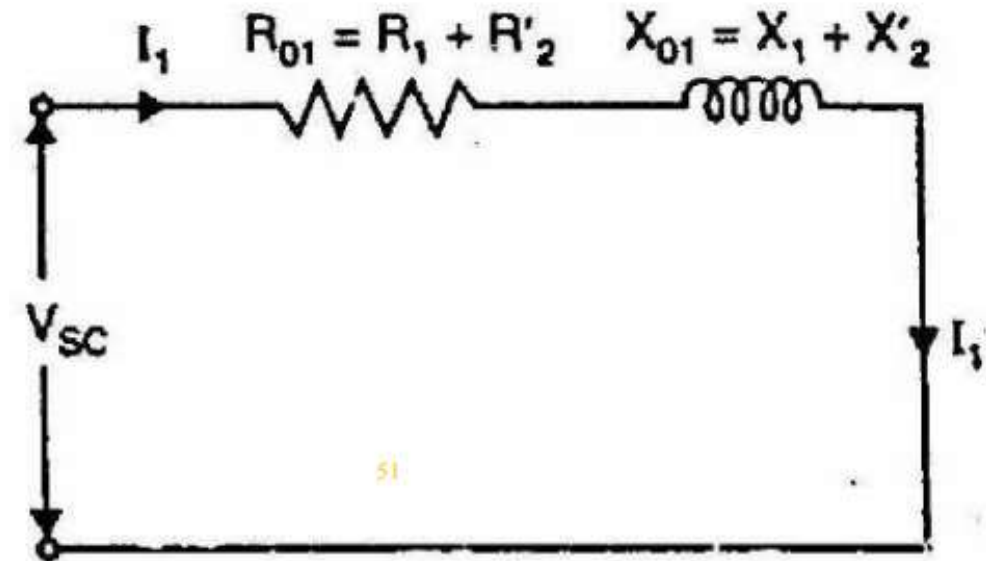
- ▶ Full load Cu loss, $P_C = \text{Wattmeter reading} = W_s$
- ▶ Applied voltage = Voltmeter reading = V_{sc}
- ▶ Full Load primary current = Ammeter reading = I_1



$$P_C = I_1^2 R_1 + I_1^2 R'_2 = I_1^2 R_{01}$$

$$\therefore R_{01} = \frac{P_C}{I_1^2}$$

R_{01} is the total resistance of transformer referred to primary.



Short Circuit Test

Total impedance referred to primary, $Z_{01} = \frac{V_{SC}}{I_1}$

Total leakage reactance referred to primary, $X_{01} = \sqrt{Z_{01}^2 - R_{01}^2}$

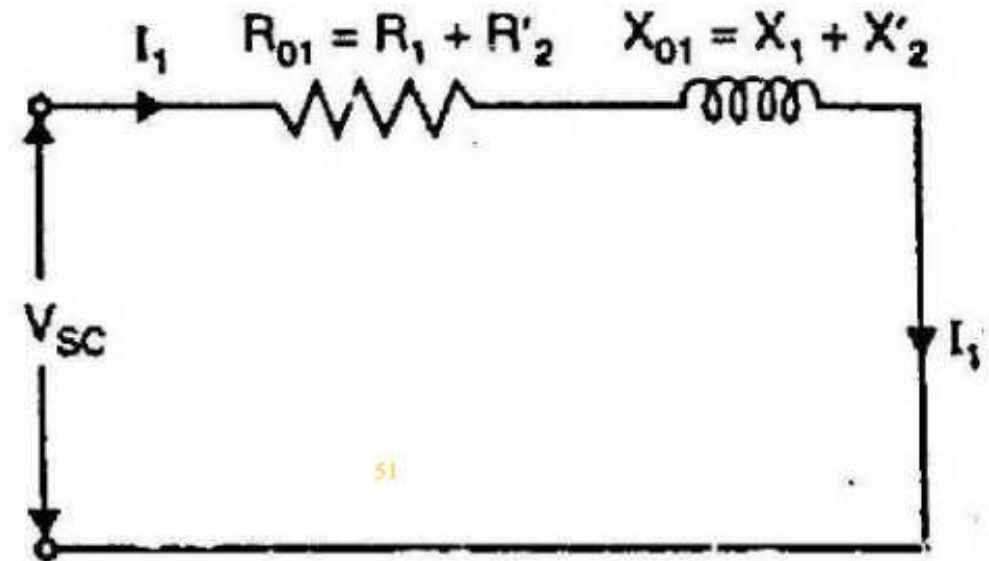
Short-circuit p.f, $\cos \phi_2 = \frac{P_C}{V_{SC} I_1}$

Thus short-circuit test gives full-load Cu loss, R_{01} and X_{01}

$$P_C = I_1^2 R_1 + I_1^2 R'_2 = I_1^2 R_{01}$$

$$\therefore R_{01} = \frac{P_C}{I_1^2}$$

R_{01} is the total resistance of transformer referred to primary.



Transformer Numerical Problems

Example. 32.27. In no-load test of single-phase transformer, the following test data were obtained :

Primary voltage : 220 V ; Secondary voltage : 110 V ;

Primary current : 0.5 A ; Power input : 30 W.

Find the following :

(i) The turns ratio (ii) the magnetising component of no-load current (iii) its working (or loss) component (iv) the iron loss.

Resistance of the primary winding = 0.6 ohm.

Draw the no-load phasor diagram to scale.

(Elect. Machine A.M.I.E. 1990)

Solution. (i) Turn ratio, $N_1/N_2 = 220/110 = 2$

(ii) $W = V_1 I_0 \cos \phi_0$; $\cos \phi_0 = 30/220 \times 0.5 = 0.273$; $\sin \phi_0 = 0.962$

$$I_\mu = I_0 \sin \phi_0 = 0.5 \times 0.962 = 0.48 \text{ A}$$

(iii) $I_w = I_0 \cos \phi_0 = 0.5 \times 0.273 = 0.1365 \text{ A}$

(iv) Primary Cu loss = $I_0^2 R_1 = 0.5^2 \times 0.6 = 0.15 \text{ W}$

$\therefore$ Iron loss = $30 - 0.15 = 29.85 \text{ W}$

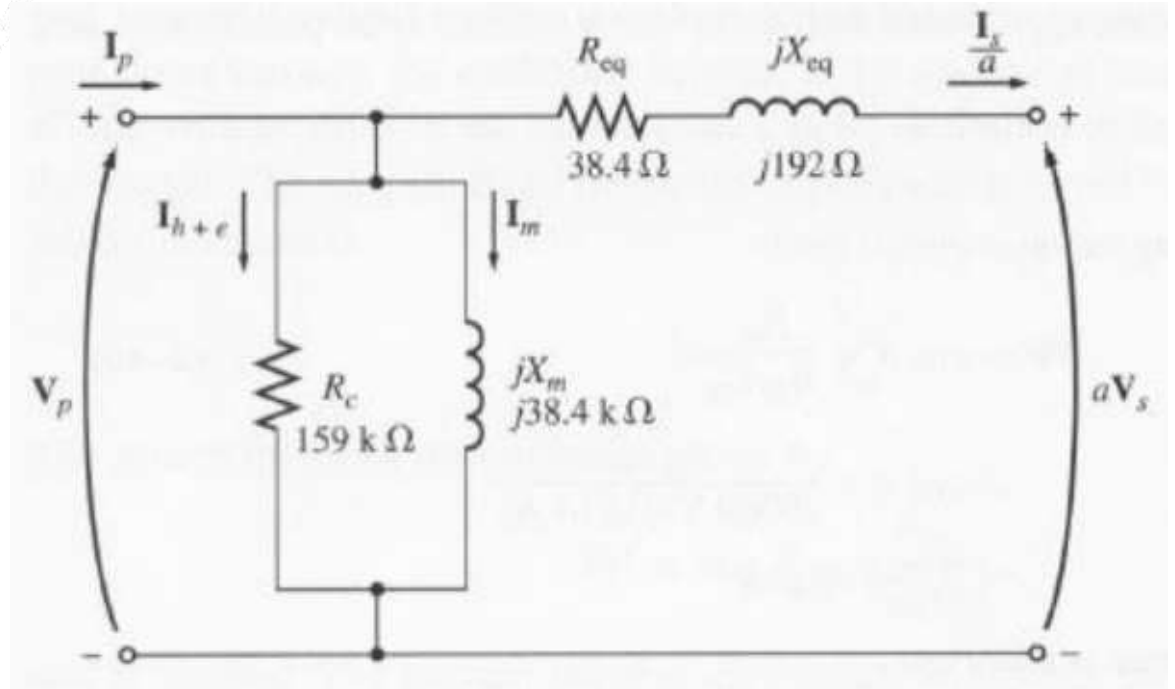
Transformer Numerical Problems

The equivalent circuit impedances of a 20-kVA, 8000/240-V, 60-Hz transformer are to be determined. The open-circuit test and the short-circuit test were performed on the primary side of the transformer, and the following data were taken:

Open-circuit test (on primary)	Short-circuit test (on primary)
$V_{oc} = 8000 \text{ V}$	$V_{sc} = 489 \text{ V}$
$I_{oc} = 0.214 \text{ A}$	$I_{sc} = 2.5 \text{ A}$
$P_{oc} = 400 \text{ W}$	$P_{sc} = 240 \text{ W}$

Find the impedances of the approximate equivalent circuit referred to the primary side, and sketch the circuit.

Previous Question - Spring 2025
Try Yourself



Transformer Numerical Problems

Example 32.32. When a transformer is connected to a 1000-V, 50-Hz supply the core loss is 1000 W, of which 650 is hysteresis and 350 is eddy current loss. If the applied voltage is raised to 2,000 V and the frequency to 100 Hz, find the new core losses.

Solution. Hysteresis loss $W_h \propto B_{\max}^{1.6} f = P B_{\max}^{1.6} f$

Eddy current loss $W_e \propto B_{\max}^2 f^2 = Q B_{\max}^2 f^2$

From the relation $E = 4.44 f N B_{\max}$ A volt, we get $B_{\max} \propto E/f$

Putting this value of $B_{\max}$ in the above equations, we have

$$W_h = P \left(\frac{E}{f} \right)^{1.6} f = P E^{1.6} f^{-0.6} \text{ and } W_e = Q \left(\frac{E}{f} \right)^2 f^2 = Q E^2$$

In the first case,

$$E = 1000 \text{ V}, f = 50 \text{ Hz}, W_h = 650 \text{ W}, W_e = 350 \text{ W}$$

$$\therefore 650 = P \times 1000^{1.6} \times 50^{-0.6} \quad \therefore P = 650 \times 1000^{-1.6} \times 50^{0.6}$$

Similarly,

$$350 = Q \times 1000^2 \quad \therefore Q = 350 \times 1000^{-2}$$

Hence, constants P and Q are known.

Using them in the second case, we get

$$W_h = (650 \times 1000^{-1.6} \times 50^{0.6}) \times 2000^{1.6} \times 100^{-0.6} = 650 \times 2 = 1,300 \text{ W}$$

$$W_e = (350 \times 1000^{-2}) \times 2,000^2 = 350 \times 4 = 1,400 \text{ W}$$

$\therefore$ Core loss under new condition is $= 1,300 + 1,400 = 2700 \text{ W}$

Alternative Solution

Here, both voltage and frequency are doubled, leaving the flux density unchanged.

With 1000 V at 50 Hz

$$W_h = A f \text{ or } 650 = 50 A; A = 13$$

$$W_e = B f^2 \text{ or } 350 = B \times 50^2; B = 7/50$$

With 2000 V at 100 Hz

$$W_h = A f = 13 \times 100 = 1300 \text{ W and}$$

$$W_e = B f^2 = (7/50) \times 100^2 = 1400 \text{ W}$$

$\therefore$ New core loss $= 1300 + 1400 = 2700 \text{ W}$

Advantages of Transformer Tests

The above two simple transformer tests offer the following advantages:

- (i) The power required to carry out these tests is very small as compared to the full-load output of the transformer.
- (ii) These tests enable us to determine the efficiency of the transformer accurately at any load and p.f. without actually loading the transformer.
- (iii) The short-circuit test enables us to determine R_{01} and X_{01} (or R_{02} and X_{02}). We can thus find the total voltage drop in the transformer as referred to primary or secondary. This permits us to calculate voltage regulation of the transformer.